

SHETH L.U.J AND SIR M.V. COLLEGE
SUBJECT NAME: **Cyber & Information Security**
Practical : 2

AIM : Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations. Using cli

#with number

```
from math import gcd
```

```
p = int(input("Enter prime p: "))
q = int(input("Enter prime q: "))
m = int(input("Enter message (number): "))
```

```
n = p * q
phi = (p - 1) * (q - 1)
```

```
e = 2
while gcd(e, phi) != 1:
    e += 1
```

```
d = pow(e, -1, phi)
```

```
c = pow(m, e, n)
msg = pow(c, d, n)
```

```
print("Public Key :", (e, n))
print("Private Key:", (d, n))
print("Encrypted :", c)
print("Decrypted :", msg)
```

#with name

```
p = int(input("Enter prime p: "))
q = int(input("Enter prime q: "))
text = input("Enter name: ")
```

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```
from math import gcd

n = p * q
phi = (p - 1) * (q - 1)

e = 2
while gcd(e, phi) != 1:
    e += 1

d = pow(e, -1, phi)

enc = [pow(ord(ch), e, n) for ch in text]
dec = "".join(chr(pow(x, d, n)) for x in enc)

print("Encrypted:", enc)
print("Decrypted:", dec)
```

Output :

```
=====
Enter prime p: 115
Enter prime q: 116
Enter message (number): 2
Public Key : (7, 13340)
Private Key: (1873, 13340)
Encrypted   : 128
Decrypted   : 5348

===== RESTART: C:/Users/IT-50/Downloads/cybersecurty practc
Enter prime p: 115
Enter prime q: 116
Enter message (number): 2
Public Key : (7, 13340)
Private Key: (1873, 13340)
Encrypted   : 128
Decrypted   : 5348
Enter prime p: 117
Enter prime q: 116
Enter name: adarsh shukla
Encrypted: [3349, 9244, 3349, 2196, 811, 11960, 5624, 811, 11960, 117, 3563, 11088, 3349, 5624]
Decrypted: 7-9:9Φ9Φ9Φu0W?9
> |
```

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Screen shots

```
from math import gcd

p = int(input("Enter prime p: "))
q = int(input("Enter prime q: "))
m = int(input("Enter message (number): "))

n = p * q
phi = (p - 1) * (q - 1)

e = 2
while gcd(e, phi) != 1:
    e += 1

d = pow(e, -1, phi)

c = pow(m, e, n)
msg = pow(c, d, n)

print("Public Key :", (e, n))
print("Private Key:", (d, n))
print("Encrypted  :", c)
print("Decrypted  :", msg)
```

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```
p = int(input("Enter prime p: "))
q = int(input("Enter prime q: "))
text = input("Enter name: ")

from math import gcd

n = p * q
phi = (p - 1) * (q - 1)

e = 2
while gcd(e, phi) != 1:
    e += 1

d = pow(e, -1, phi)

enc = [pow(ord(ch), e, n) for ch in text]
dec = "".join(chr(pow(x, d, n)) for x in enc)

print("Encrypted:", enc)
print("Decrypted:", dec)
```

Output

```
C:\Users\IT-50>"C:\Users\IT-50\Downloads\cybersecurity practcail 2.py"
Enter prime p: 116
Enter prime q: 117
Enter message (number): 12
Public Key : (3, 13572)
Private Key: (4447, 13572)
Encrypted   : 1728
Decrypted   : 4536
Enter prime p: 117
Enter prime q: 116
Enter name: adarsh shukla
Encrypted: [3349, 9244, 3349, 2196, 811, 11960, 5624, 811, 11960, 117, 3
563, 11088, 3349]
Decrypted: 7---7::ē φÄē φuüŴ?
```

Aim : Implement the RSA algorithm for public-key encryption and decryption, and explore its properties and security considerations. Using gui

Code :

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```
import tkinter as tk
from tkinter import messagebox
from math import gcd

# ----- RSA LOGIC -----
def generate_keys(p, q):
    n = p * q
    phi = (p - 1) * (q - 1)

    e = next(e for e in range(2, phi) if gcd(e, phi) == 1)
    d = pow(e, -1, phi)

    return e, d, n

def rsa_process(p, q, data, mode):
    e, d, n = generate_keys(p, q)

    if mode == "Number":
        data = int(data)
        enc = pow(data, e, n)
        dec = pow(enc, d, n)

    else: # Text
        enc = [pow(ord(ch), e, n) for ch in data]
        dec = "".join(chr(pow(x, d, n)) for x in enc)

    return e, d, n, enc, dec

# ----- GUI -----
def process():
    try:
        p, q = int(entry_p.get()), int(entry_q.get())
        data = entry_data.get()
        mode = mode_var.get()

        e, d, n, enc, dec = rsa_process(p, q, data, mode)

        output.set(
            f"Public Key : ({e}, {n})\n"
            f"Private Key: ({d}, {n})\n"
```

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```
f"Encrypted : {enc}\n"
f"Decrypted : {dec}"
)

except Exception:
    messagebox.showerror("Error", "Invalid input")

# ----- WINDOW -----
root = tk.Tk()
root.title("RSA GUI")
root.geometry("380x350")
root.resizable(False, False)

# Title
tk.Label(root, text="RSA Encryption", font=("Arial", 14, "bold")).pack(pady=10)

# Inputs
frame = tk.Frame(root)
frame.pack(pady=5)

tk.Label(frame, text="p:").grid(row=0, column=0)
entry_p = tk.Entry(frame, width=10)
entry_p.grid(row=0, column=1)

tk.Label(frame, text="q:").grid(row=1, column=0)
entry_q = tk.Entry(frame, width=10)
entry_q.grid(row=1, column=1)

tk.Label(frame, text="Input:").grid(row=2, column=0)
entry_data = tk.Entry(frame, width=20)
entry_data.grid(row=2, column=1)

# Mode
mode_var = tk.StringVar(value="Number")
tk.OptionMenu(root, mode_var, "Number", "Text").pack(pady=5)

# Button
tk.Button(root, text="Run", command=process, bg="black", fg="white", width=15).pack(pady=10)

# Output
output = tk.StringVar()
```

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```
tk.Label(root, textvariable=output, bg="#eee", width=45, height=8, justify="left").pack(pady=10)
```

```
# Run  
root.mainloop()
```

Screen shots code gui

```
import tkinter as tk
from tkinter import messagebox
from math import gcd

# ----- RSA LOGIC -----
def generate_keys(p, q):
    n = p * q
    phi = (p - 1) * (q - 1)

    e = next(e for e in range(2, phi) if gcd(e, phi) == 1)
    d = pow(e, -1, phi)

    return e, d, n

def rsa_process(p, q, data, mode):
    e, d, n = generate_keys(p, q)

    if mode == "Number":
        data = int(data)
        enc = pow(data, e, n)
        dec = pow(enc, d, n)

    else: # Text
        enc = [pow(ord(ch), e, n) for ch in data]
        dec = "".join(chr(pow(x, d, n)) for x in enc)

    return e, d, n, enc, dec

# ----- GUI -----
def process():
    try:
        p, q = int(entry_p.get()), int(entry_q.get())
        data = entry_data.get()
        mode = mode_var.get()

        e, d, n, enc, dec = rsa_process(p, q, data, mode)

        output.set(
            f"Public Key : ({e}, {n})\n"
            f"Private Key: ({d}, {n})\n"
        )
    except ValueError:
        messagebox.showerror("Error", "Invalid input")
```

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```
# ----- GUI -----
def process():
    try:
        p, q = int(entry_p.get()), int(entry_q.get())
        data = entry_data.get()
        mode = mode_var.get()

        e, d, n, enc, dec = rsa_process(p, q, data, mode)

        output.set(
            f"Public Key : {(e), {n}}\n"
            f"Private Key: {(d), {n}}\n"
            f"Encrypted  : {enc}\n"
            f"Decrypted   : {dec}"
        )

    except Exception:
        messagebox.showerror("Error", "Invalid input")

# ----- WINDOW -----
root = tk.Tk()
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entry_p = tk.Entry(frame, width=10)
entry_p.grid(row=0, column=1)
```

```
# ----- WINDOW -----
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# Inputs
frame = tk.Frame(root)
frame.pack(pady=5)

tk.Label(frame, text="p:") .grid(row=0, column=0)
entry_p = tk.Entry(frame, width=10)
entry_p.grid(row=0, column=1)

tk.Label(frame, text="q:") .grid(row=1, column=0)
entry_q = tk.Entry(frame, width=10)
entry_q.grid(row=1, column=1)

tk.Label(frame, text="Input:") .grid(row=2, column=0)
entry_data = tk.Entry(frame, width=20)
entry_data.grid(row=2, column=1)

# Mode
mode_var = tk.StringVar(value="Number")
tk.OptionMenu(root, mode_var, "Number", "Text").pack(pady=5)

# Button
tk.Button(root, text="Run", command=process, bg="black", fg="white", width=15).pack(pady=10)

# Output
output = tk.StringVar()
tk.Label(root, textvariable=output, bg="#eee", width=45, height=8, justify="left").pack(pady=10)

# Run
root.mainloop()
```

Output :

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RSA Encryption

p:

q:

Input:

3, 13572)
4447, 13572)
3349, 9244, 3349, 2196, 811, 11960, 5624, 811, 11960, 117, 3563,
7-7::ē+Äē+u∩W?

RSA GUI

RSA Encryption

p:

q:

Input:

3, 13572)
4447, 13572)
3349, 9244, 3349
7-7::ē+Äē+u∩W?

1960, 117, 3!

Error

Invalid input

OK

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RSA Encryption

p:

q:

Input:

Public Key : (3, 13572)
Private Key: (4447, 13572)
Encrypted : 944
Decrypted : 11276

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